

## 2.5.3 Operations on Data Structures

بر ڈیٹا سٹرکچر پر پانچ عمل ہوتے ہیں: داخل کرنا (Insertion)، حذف کرنا (Deletion)، ایک ایک کر کے گزرنا (Traversa)...

CS XII

Chapter 2: Computational Thinking &amp; Algorithms

Lecture 21 of 24

### 2.5.3 Operations on Data Structures

#### LEARN IT

Whatever the shape, you keep meeting the same five operations:

| Operation | Meaning                                                      | Quick picture                                                  |
|-----------|--------------------------------------------------------------|----------------------------------------------------------------|
| Insertion | Adding a new item into the structure.                        | PUSH on a stack; ENQUEUE on a queue; putting 40 into an array. |
| Deletion  | Removing an item that is no longer needed.                   | POP; DEQUEUE; deleting a node from a list.                     |
| Traversal | Going through each item one by one.                          | Walking HEAD → ... → NULL; visiting every tree node.           |
| Searching | Finding a specific item.                                     | Linear search in an array; following links until 87 is found.  |
| Sorting   | Arranging items in a proper order (ascending or descending). | Bubble Sort on an array of marks.                              |

#### WORKED EXAMPLE

**Worked example: name the operation.**

1. "Print every song from the start of the playlist." → **Traversal**.
2. "Add a print job at the rear." → **Insertion** (ENQUEUE).
3. "Undo the last typing action." → **Deletion** (POP).
4. "Is roll number 87 in the list?" → **Searching**.
5. "Arrange heights from shortest to tallest." → **Sorting**.

#### REMEMBER IT

Memorise **I-D-T-S-S**: Insert, Delete, Traverse, Search, Sort.

#### COMMON MISTAKE

**Mistake:** calling a PUSH "sorting" because the stack looks tidy.

**Fix:** PUSH is insertion. Sorting means putting values in ascending or descending order.

اردو میں سمجھیں

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#### CHECK YOURSELF

1. List the five operations.
2. Which operation is ENQUEUE?
3. Traversal vs searching — one difference?

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Answers: 1. Insertion, deletion, traversal, searching, sorting. 2. Insertion (at the rear of a queue). 3. Traversal visits every item; searching stops when the target is found (or the structure ends).

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Next lecture: 2.6 Identify data structures in problem scenarios